Published-bearing case: Cape Leeuwin 00:49:58 and Diego Garcia South 01:10:15 as one impact near 35.9° S, 92.7° E

Candidate: Cape Leeuwin 00:49:58 (Kadri 2024 Table 1; published bearing 260.4°, the sensitivity case under ruling B1) with the Diego Garcia South envelope peak 01:10:15 as one impact. Joint solution (4,000 Monte Carlo draws, physical speed bounds): source 35.9° S 92.7° E; 50 % ellipse 42 x 25 NM, 95 % ellipse 87 x 51 NM (long axis 76°); 4-78 NM from the 7th arc near 34.9-36.0° S; impact 00:26:36-00:28:49 UTC (90 %); mean ground speed needed from the arc 31-542 kt (90 %).

Site	Path	Expected arrival, physical bounds (95 %)	Expected arrival, Kadri speeds (95 %)	Recorded data in window	Window p (combined)	Best look (p)	Events noted earlier: time, position in expected range (physical / Kadri)
Cape Leeuwin	clear	00:49:38–00:49:57	00:49:38–00:49:57	Kadri trace	—	defining event	defines the candidate
Diego Garcia South	clear, ridge (partial)	01:09:55–01:10:14	01:09:55–01:10:14	Kadri trace	—	defining event	defines the candidate
Scott Reef	blocked (all draws)	01:05:52–01:09:48	01:06:29–01:10:21	0 %	—	—	00:44:56: 0 / 0 %; 00:48:49: 0 / 0 %; 01:16:34: 100 / 100 %; 01:32:51: 100 / 100 %; 01:04:30: 0 / 0 %
Portland 1	clear	01:13:42–01:18:22	01:13:19–01:17:39	72 %	0.789	01:15:59 (0.229)	01:00:39: 0 / 0 %; 01:17:37: 88 / 97 % (inside); 01:32:33: 100 / 100 %
Perth Canyon 1	clear	00:50:07–00:53:05	00:50:13–00:52:58	100 %	0.733	00:53:07 (0.245)	00:35:42: 0 / 0 %
Perth Canyon 2	clear	00:50:06–00:53:05	00:50:10–00:52:55	1 %	0.987	00:49:51 (0.911)	01:02:10: 100 / 100 %; 00:46:31: 0 / 0 %
Portland 2	clear	01:13:42–01:18:21	01:13:14–01:17:34	0 %	—	—	01:07:44: 0 / 0 %; 01:20:30: 100 / 100 %

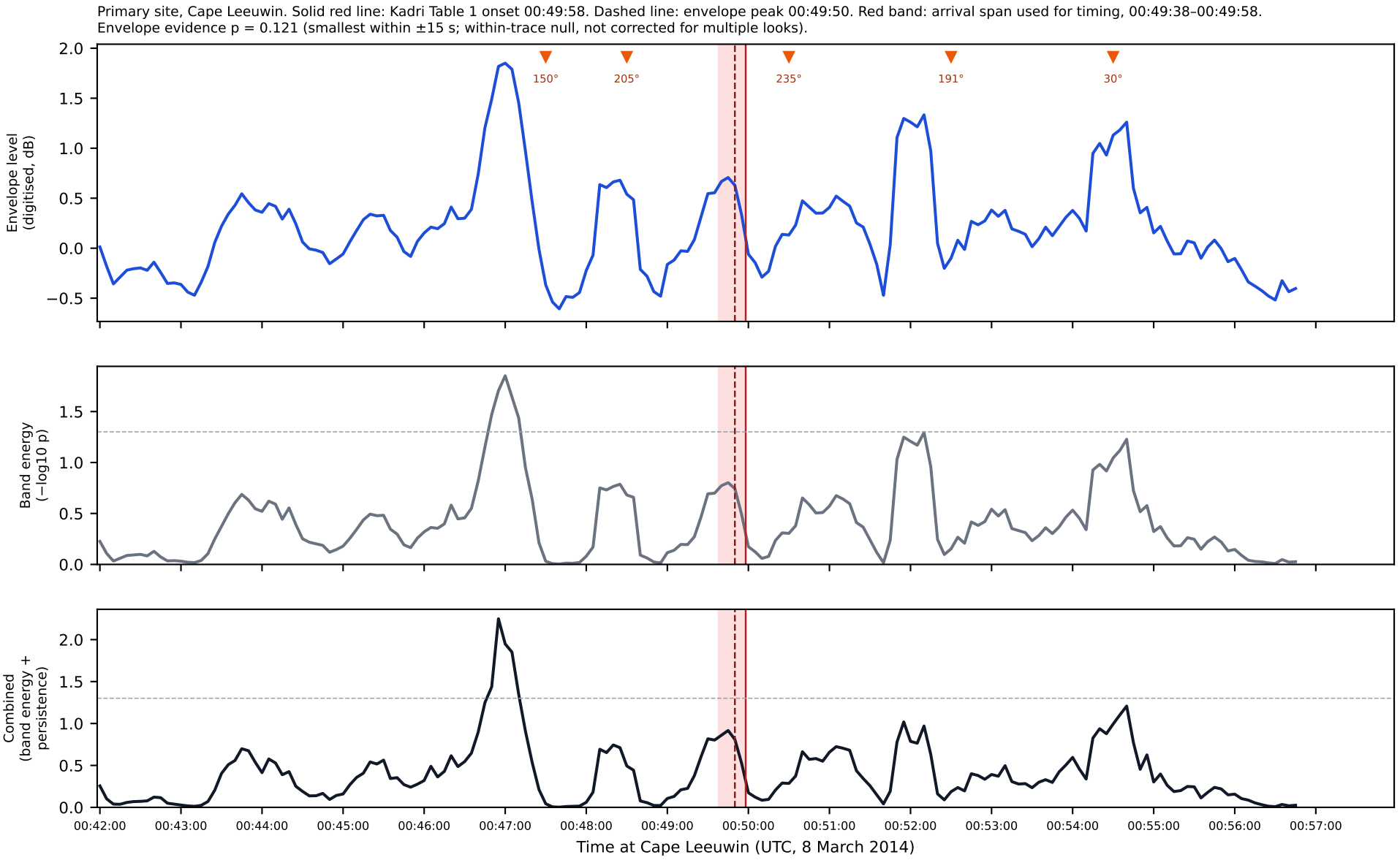
Position in expected range: percentile of the noted event time within the predicted arrival distribution (0 % = earlier than every draw, 100 % = later than every draw); 'inside' = within the 95 % range. Events noted earlier come from the Cape Leeuwin / Diego Garcia South candidate questions of 7 Nov 2026 (Scott Reef low-frequency band 00:45-00:48; Portland 1 just before 01:01, about 01:15, about 01:33; Portland 2 01:07 and about 01:21; Perth Canyon 1 correlation 00:35-00:36), the Scott Reef picks in Kadri, Crivelli and Parsons (2017) ATSB report Fig. 6 (logger time corrected by -15.1 s), and the best combined looks of earlier candidate windows. The published bearing is the sensitivity case; with the corrected bearing (246.8°, primary under ruling B1) the source lies 122-202 NM from the arc and needs at least 849 kt mean ground speed in every draw. Recorded data in window: share of the physical 95 % range covered by logger records. Window p uses the widened window on each site page and is not corrected for the number of candidates and sites examined. Cape Leeuwin and Diego Garcia South: digitised Kadri (2024) Figure 9 envelopes only. Bearing from Diego Garcia South to this source: 147.8-151.6° (95 %); for the corrected-bearing source 154.8-159.9°. Preliminary, not checked: the other thread's digitisation of Kadri (2019) Fig. 4 (in progress, not yet reported) lists for 01:09:34-01:10:18 a dominant bearing of 311° (the shot-train direction of Kadri 2019 Fig. 1) and a strongest non-shot bearing of 39-42°; neither is near 150°. This page will be updated when that work is reported.



Wide grey bar: expected arrival, physical bounds (95 %). Thin blue bar: Kadri speeds (95 %).
Black segments under each row: IMOS logger records. Orange triangles: events noted earlier (label only).

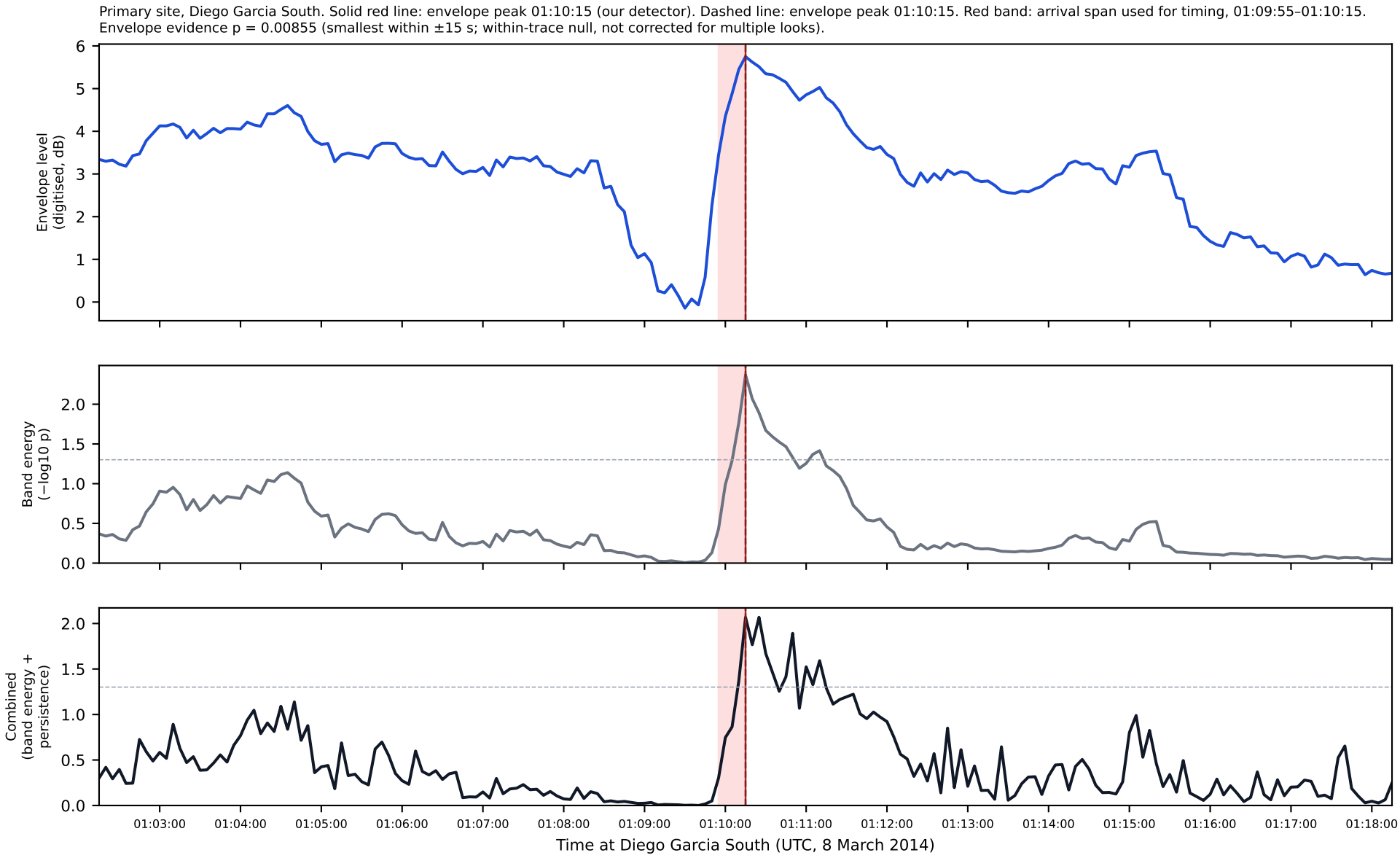
Candidate: Cape Leeuwin 00:49:58 (Kadri 2024 Table 1; published bearing 260.4°, the sensitivity case under ruling B1) with the Diego Garcia South envelope peak 01:10:15 as one impact. Joint solution (4,000 Monte Carlo draws, physical speed bounds): source 35.9° S 92.7° E; 50 % ellipse 42 x 25 NM, 95 % ellipse 87 x 51 NM (long axis 76°); 4-78 NM from the 7th arc near 34.9-36.0° S; impact 00:26:36-00:28:49 UTC (90 %); mean ground speed needed from the arc 31-542 kt (90 %). Cape Leeuwin and Diego Garcia South rows show the arrival spans that define the candidate. Speeds. Grey shading and wide grey bars: physical bounds for sound in the water on each path (WOA23 March climatology, 1°, sampled every 25 km on the great circle; slowest = path mean of the sound-channel axis speed minus 2 sd, then 5 % longer path; fastest = path mean of the fastest column speed plus 2 sd). Blue dashed lines and thin blue bars: Kadri (2024) 1.45-1.57 km/s, as published. Both are 95 % ranges over 4,000 Monte Carlo draws of the joint source (published Cape Leeuwin bearing 260.4°, sd 1.26°; arrival spans onset minus 20 s to envelope peak; no grid and no reachability cap). Window shown: both ranges together, widened by 20 % at each end. Path class from the GEBCO 2026 screen at the nearest 25 NM grid point. IMOS loggers record 307 s in every 15 min (Portland 1 every 10 min); grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), calibrated on 14 days of noise (1-15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15-02:00; band energy 8-40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of five scores. Calibration: USGS M≥3.5 earthquake arrival windows and faulty records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum). Window p: chance that noise alone gives a maximum this high over the looks in the window. Each score uses 30 s windows and a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median). Orange ticks: events noted earlier at this site (label only). Top strip: expected arrival for each nearest arc latitude (0.25° bins).

Published-bearing case: Cape Leeuwin 00:49:58 (260.4° published) with Diego Garcia South 01:10:15. Source 35.9° S 92.7° E, impact 00:26:36–00:28:49 UTC



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is held for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset minus 20 s to the envelope peak (project convention of 7 Nov 2026). The p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05. Orange triangles: Cape Leeuwin detections in Kadri, Crivelli and Parsons (2017) ATSB report Fig. 3 (minute resolution, drawn at the minute centre) with their published bearings (label only). Our check does not reconcile this list with the 2024 Table 1 except at 00:50; cause not known; it can be our error. Candidate: Cape Leeuwin 00:49:58 (Kadri 2024 Table 1; published bearing 260.4°, the sensitivity case under ruling B1) with the Diego Garcia South envelope peak 01:10:15 as one impact. Joint solution (4,000 Monte Carlo draws, physical speed bounds): source 35.9° S 92.7° E; 50 % ellipse 42 x 25 NM, 95 % ellipse 87 x 51 NM (long axis 76°); 4-78 NM from the 7th arc near 34.9-36.0° S; impact 00:26:36-00:28:49 UTC (90 %); mean ground speed needed from the arc 31-542 kt (90 %).

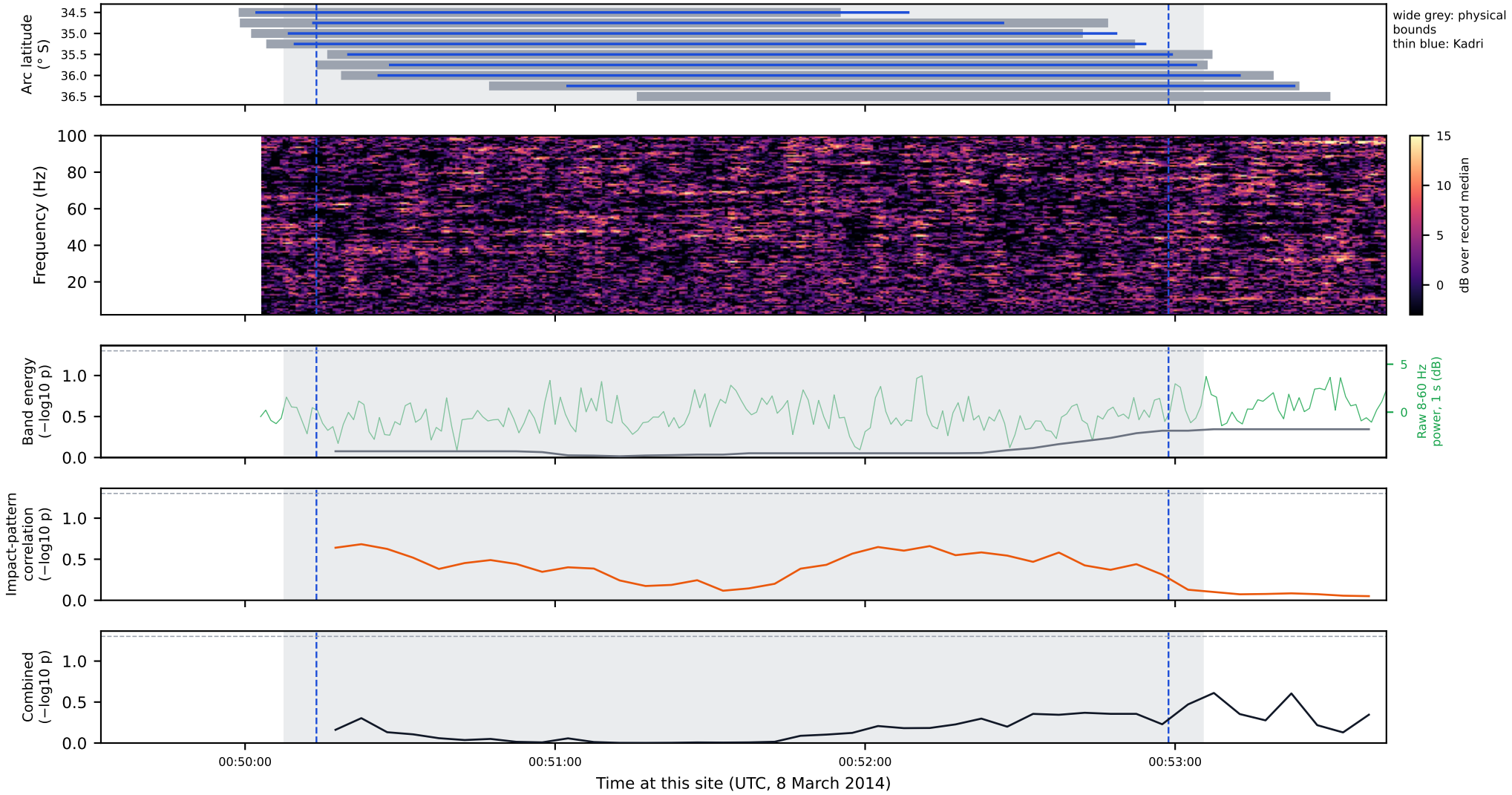
Published-bearing case: Cape Leeuwin 00:49:58 (260.4° published) with Diego Garcia South 01:10:15. Source 35.9° S 92.7° E, impact 00:26:36–00:28:49 UTC



Primary-site page. Data: digitised envelope of Kadri (2024) Figure 9 (no raw waveform is held for 8 March 2014). Band energy: level over the local background, scored against the rest of the same trace. Combined: band energy and persistence (Fisher). Arrival span: onset minus 20 s to the envelope peak (project convention of 7 Nov 2026). The p floor is about 1/355 at Cape Leeuwin and 1/234 at Diego Garcia South. Grey dashed line: p = 0.05. Bearing from Diego Garcia South to this source: 147.8-151.6° (95 %); for the corrected-bearing source 154.8-159.9°. Preliminary, not checked: the other thread's digitisation of Kadri (2019) Fig. 4 (in progress, not yet reported) lists for 01:09:34-01:10:18 a dominant bearing of 311° (the shot-train direction of Kadri 2019 Fig. 1) and a strongest non-shot bearing of 39-42°; neither is near 150°. This page will be updated when that work is reported. Candidate: Cape Leeuwin 00:49:58 (Kadri 2024 Table 1; published bearing 260.4°, the sensitivity case under ruling B1) with the Diego Garcia South envelope peak 01:10:15 as one impact. Joint solution (4,000 Monte Carlo draws, physical speed bounds): source 35.9° S 92.7° E; 50 % ellipse 42 x 25 NM, 95 % ellipse 87 x 51 NM (long axis 76°); 4-78 NM from the 7th arc near 34.9-36.0° S; impact 00:26:36-00:28:49 UTC (90 %); mean ground speed needed from the arc 31-542 kt (90 %).

Published-bearing case: Cape Leeuwin 00:49:58 (260.4° published) with Diego Garcia South 01:10:15. Source 35.9° S 92.7° E, impact 00:26:36–00:28:49 UTC

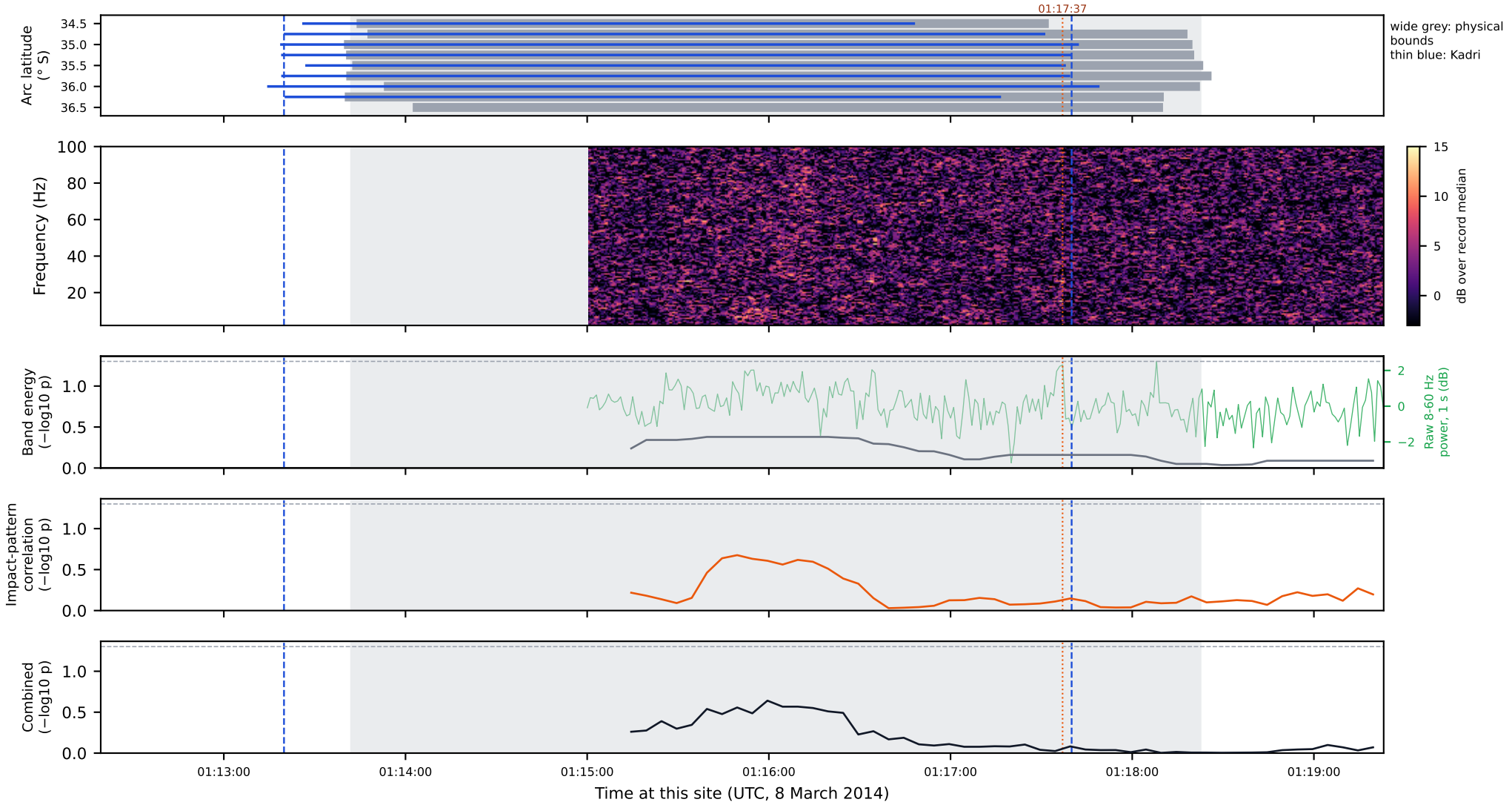
Perth Canyon 1 (IMOS 3376). Expected arrival: physical bounds 00:50:07–00:53:05 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:50:13–00:52:58 (blue dashed lines, thin blue bars).
Window shown 00:49:32–00:53:40 (20 % margin). Path: clear; 41 looks; window p (combined) = 0.733.



Speeds. Grey shading and wide grey bars: physical bounds for sound in the water on each path (WOA23 March climatology, 1°, sampled every 25 km on the great circle; slowest = path mean of the sound-channel axis speed minus 2 sd, then 5 % longer path; fastest = path mean of the fastest column speed plus 2 sd). Blue dashed lines and thin blue bars: Kadri (2024) 1.45-1.57 km/s, as published. Both are 95 % ranges over 4,000 Monte Carlo draws of the joint source (published Cape Leeuwin bearing 260.4°, sd 1.26°; arrival spans onset minus 20 s to envelope peak; no grid and no reachability cap). Window shown: both ranges together, widened by 20 % at each end. Path class from the GEBCO 2026 screen at the nearest 25 NM grid point. IMOS loggers record 307 s in every 15 min (Portland 1 every 10 min); grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), calibrated on 14 days of noise (1-15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15-02:00; band energy 8-40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of five scores. Calibration: USGS M≥3.5 earthquake arrival windows and faulty records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum). Window p: chance that noise alone gives a maximum this high over the looks in the window. Each score uses 30 s windows and a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median). Orange ticks: events noted earlier at this site (label only). Top strip: expected arrival for each nearest arc latitude (0.25° bins).
Candidate: Cape Leeuwin 00:49:58 (Kadri 2024 Table 1; published bearing 260.4°, the sensitivity case under ruling B1) with the Diego Garcia South envelope peak 01:10:15 as one impact. Joint solution (4,000 Monte Carlo draws, physical speed bounds): source 35.9° S 92.7° E; 50 % ellipse 42 x 25 NM, 95 % ellipse 87 x 51 NM (long axis 76°); 4-78 NM from the 7th arc near 34.9-36.0° S; impact 00:26:36-00:28:49 UTC (90 %); mean ground speed needed from the arc 31-542 kt (90 %).

Published-bearing case: Cape Leeuwin 00:49:58 (260.4° published) with Diego Garcia South 01:10:15. Source 35.9° S 92.7° E, impact 00:26:36–00:28:49 UTC

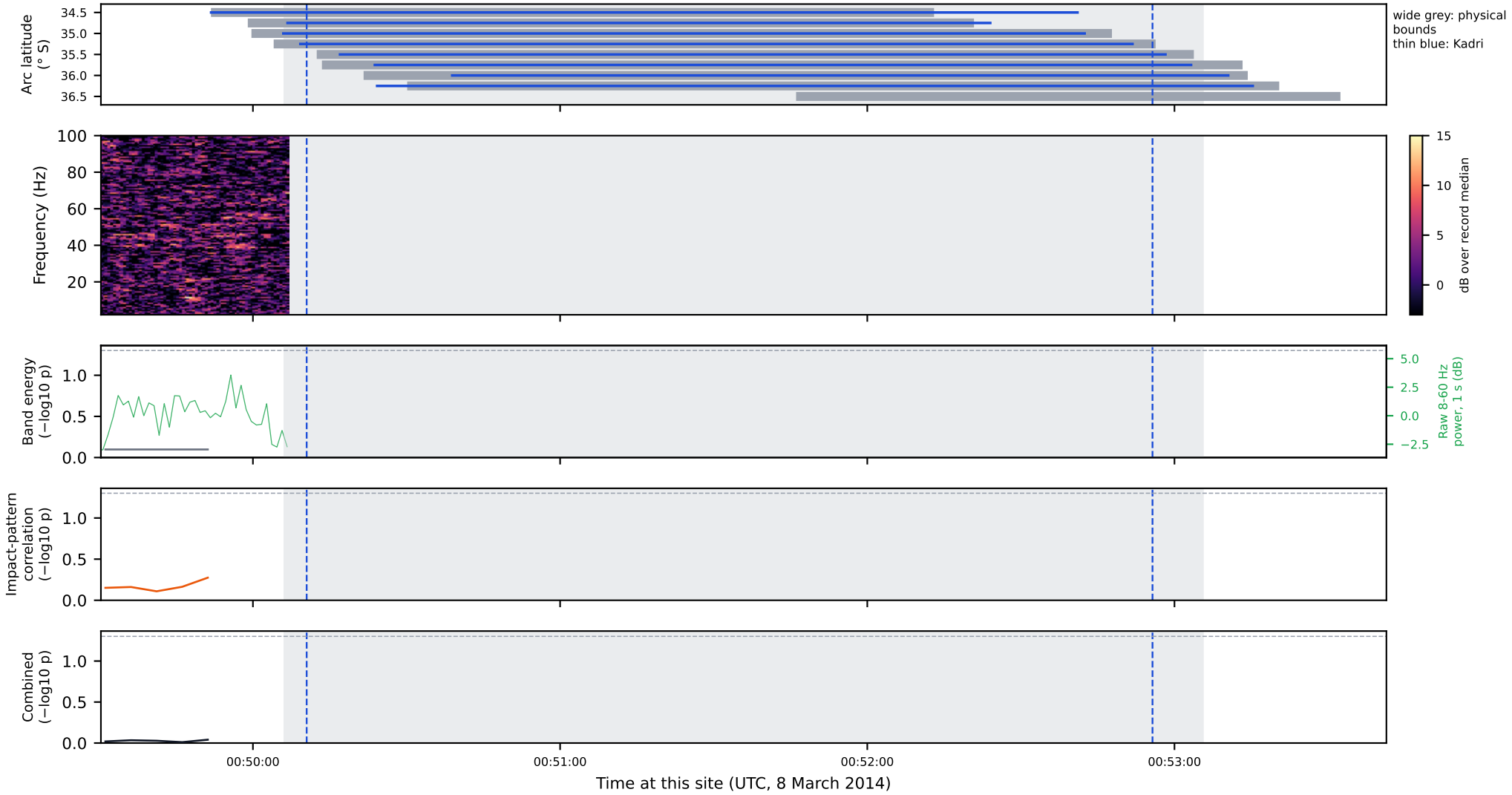
Portland 1 (IMOS 3274). Expected arrival: physical bounds 01:13:42–01:18:22 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 01:13:19–01:17:39 (blue dashed lines, thin blue bars).
Window shown 01:12:19–01:19:23 (20 % margin). Path: clear; 50 looks; window p (combined) = 0.789. Events noted earlier in view (orange dotted, position in expected range): 01:17:37 (88 %).



Speeds. Grey shading and wide grey bars: physical bounds for sound in the water on each path (WOA23 March climatology, 1°, sampled every 25 km on the great circle; slowest = path mean of the sound-channel axis speed minus 2 sd, then 5 % longer path; fastest = path mean of the fastest column speed plus 2 sd). Blue dashed lines and thin blue bars: Kadri (2024) 1.45-1.57 km/s, as published. Both are 95 % ranges over 4,000 Monte Carlo draws of the joint source (published Cape Leeuwin bearing 260.4°, sd 1.26°; arrival spans onset minus 20 s to envelope peak; no grid and no reachability cap). Window shown: both ranges together, widened by 20 % at each end. Path class from the GEBCO 2026 screen at the nearest 25 NM grid point. IMOS loggers record 307 s in every 15 min (Portland 1 every 10 min); grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), calibrated on 14 days of noise (1-15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15-02:00; band energy 8-40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of five scores. Calibration: USGS M≥3.5 earthquake arrival windows and faulty records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum). Window p: chance that noise alone gives a maximum this high over the looks in the window. Each score uses 30 s windows and a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median). Orange ticks: events noted earlier at this site (label only). Top strip: expected arrival for each nearest arc latitude (0.25° bins).
Candidate: Cape Leeuwin 00:49:58 (Kadri 2024 Table 1; published bearing 260.4°, the sensitivity case under ruling B1) with the Diego Garcia South envelope peak 01:10:15 as one impact. Joint solution (4,000 Monte Carlo draws, physical speed bounds): source 35.9° S 92.7° E; 50 % ellipse 42 x 25 NM, 95 % ellipse 87 x 51 NM (long axis 76°); 4-78 NM from the 7th arc near 34.9-36.0° S; impact 00:26:36-00:28:49 UTC (90 %); mean ground speed needed from the arc 31-542 kt (90 %).

Published-bearing case: Cape Leeuwin 00:49:58 (260.4° published) with Diego Garcia South 01:10:15. Source 35.9° S 92.7° E, impact 00:26:36–00:28:49 UTC

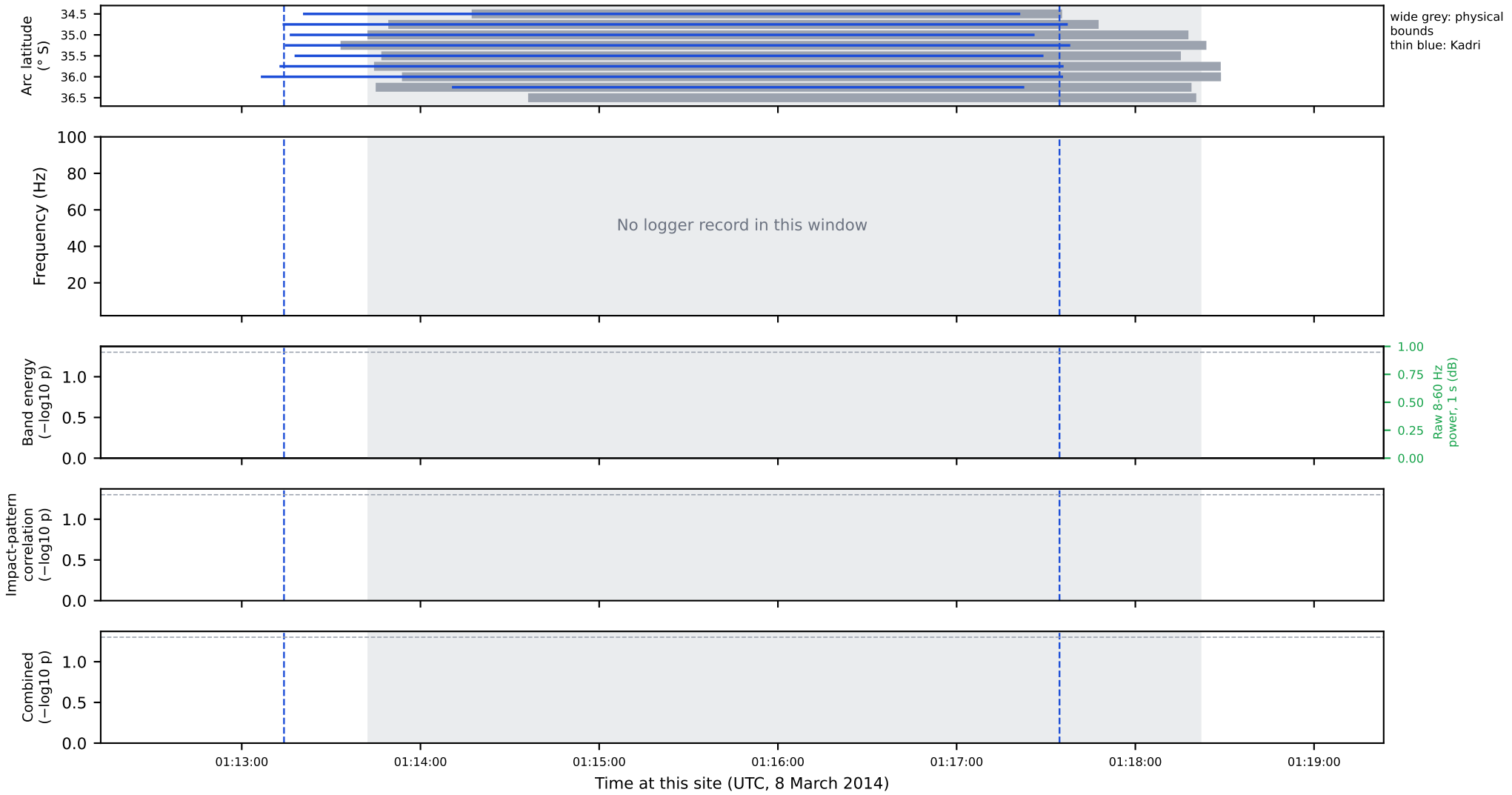
Perth Canyon 2 (IMOS 3315). Expected arrival: physical bounds 00:50:06–00:53:05 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 00:50:10–00:52:55 (blue dashed lines, thin blue bars).
Window shown 00:49:30–00:53:41 (20 % margin). Path: clear; 5 looks; window p (combined) = 0.987.



Speeds. Grey shading and wide grey bars: physical bounds for sound in the water on each path (WOA23 March climatology, 1°, sampled every 25 km on the great circle; slowest = path mean of the sound-channel axis speed minus 2 sd, then 5 % longer path; fastest = path mean of the fastest column speed plus 2 sd). Blue dashed lines and thin blue bars: Kadri (2024) 1.45-1.57 km/s, as published. Both are 95 % ranges over 4,000 Monte Carlo draws of the joint source (published Cape Leeuwin bearing 260.4°, sd 1.26°; arrival spans onset minus 20 s to envelope peak; no grid and no reachability cap). Window shown: both ranges together, widened by 20 % at each end. Path class from the GEBCO 2026 screen at the nearest 25 NM grid point. IMOS loggers record 307 s in every 15 min (Portland 1 every 10 min); grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), calibrated on 14 days of noise (1-15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15-02:00; band energy 8-40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of five scores. Calibration: USGS M≥3.5 earthquake arrival windows and faulty records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum). Window p: chance that noise alone gives a maximum this high over the looks in the window. Each score uses 30 s windows and a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median). Orange ticks: events noted earlier at this site (label only). Top strip: expected arrival for each nearest arc latitude (0.25° bins).
Candidate: Cape Leeuwin 00:49:58 (Kadri 2024 Table 1; published bearing 260.4°, the sensitivity case under ruling B1) with the Diego Garcia South envelope peak 01:10:15 as one impact. Joint solution (4,000 Monte Carlo draws, physical speed bounds): source 35.9° S 92.7° E; 50 % ellipse 42 x 25 NM, 95 % ellipse 87 x 51 NM (long axis 76°); 4-78 NM from the 7th arc near 34.9-36.0° S; impact 00:26:36-00:28:49 UTC (90 %); mean ground speed needed from the arc 31-542 kt (90 %).

Published-bearing case: Cape Leeuwin 00:49:58 (260.4° published) with Diego Garcia South 01:10:15. Source 35.9° S 92.7° E, impact 00:26:36–00:28:49 UTC

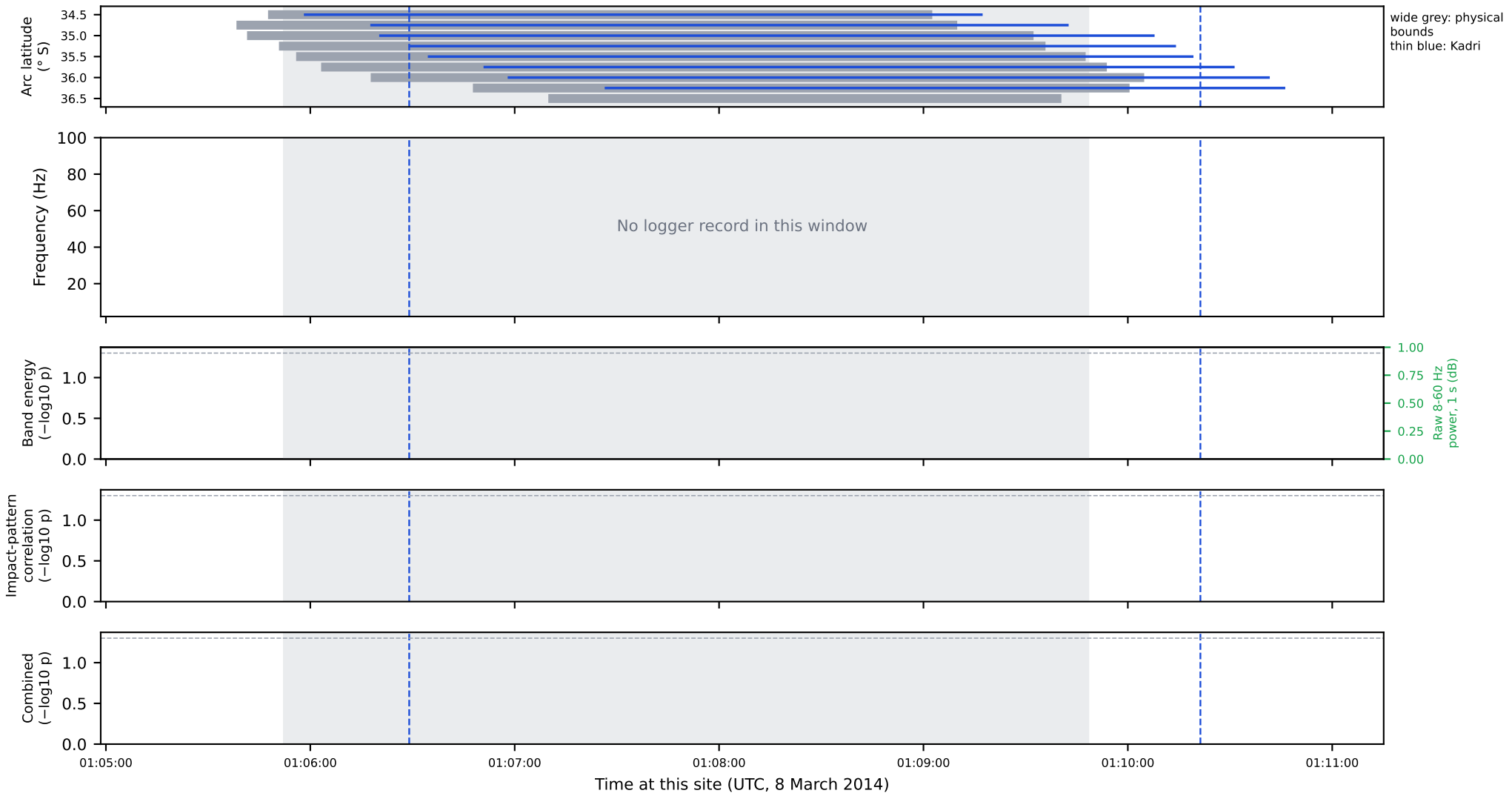
Portland 2 (IMOS 3275). Expected arrival: physical bounds 01:13:42–01:18:21 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 01:13:14–01:17:34 (blue dashed lines, thin blue bars).
Window shown 01:12:12–01:19:23 (20 % margin). Path: clear; 0 looks; window p (combined) = no data.



Speeds. Grey shading and wide grey bars: physical bounds for sound in the water on each path (WOA23 March climatology, 1°, sampled every 25 km on the great circle; slowest = path mean of the sound-channel axis speed minus 2 sd, then 5 % longer path; fastest = path mean of the fastest column speed plus 2 sd). Blue dashed lines and thin blue bars: Kadri (2024) 1.45-1.57 km/s, as published. Both are 95 % ranges over 4,000 Monte Carlo draws of the joint source (published Cape Leeuwin bearing 260.4°, sd 1.26°; arrival spans onset minus 20 s to envelope peak; no grid and no reachability cap). Window shown: both ranges together, widened by 20 % at each end. Path class from the GEBCO 2026 screen at the nearest 25 NM grid point. IMOS loggers record 307 s in every 15 min (Portland 1 every 10 min); grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), calibrated on 14 days of noise (1-15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15-02:00; band energy 8-40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of five scores. Calibration: USGS M≥3.5 earthquake arrival windows and faulty records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum). Window p: chance that noise alone gives a maximum this high over the looks in the window. Each score uses 30 s windows and a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median). Orange ticks: events noted earlier at this site (label only). Top strip: expected arrival for each nearest arc latitude (0.25° bins).
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Published-bearing case: Cape Leeuwin 00:49:58 (260.4° published) with Diego Garcia South 01:10:15. Source 35.9° S 92.7° E, impact 00:26:36–00:28:49 UTC

Scott Reef (IMOS 3250). Expected arrival: physical bounds 01:05:52–01:09:48 (grey shading, wide grey bars); Kadri 1.45–1.57 km/s 01:06:29–01:10:21 (blue dashed lines, thin blue bars).
Window shown 01:04:58–01:11:15 (20 % margin). Path: blocked by seabed in all draws; 0 looks; window p (combined) = no data.



Speeds. Grey shading and wide grey bars: physical bounds for sound in the water on each path (WOA23 March climatology, 1°, sampled every 25 km on the great circle; slowest = path mean of the sound-channel axis speed minus 2 sd, then 5 % longer path; fastest = path mean of the fastest column speed plus 2 sd). Blue dashed lines and thin blue bars: Kadri (2024) 1.45-1.57 km/s, as published. Both are 95 % ranges over 4,000 Monte Carlo draws of the joint source (published Cape Leeuwin bearing 260.4°, sd 1.26°; arrival spans onset minus 20 s to envelope peak; no grid and no reachability cap). Window shown: both ranges together, widened by 20 % at each end. Path class from the GEBCO 2026 screen at the nearest 25 NM grid point. IMOS loggers record 307 s in every 15 min (Portland 1 every 10 min); grey gaps have no data. Spectrogram: 2 s segments, power over the median of the same record at each frequency (dB). Lower panels: chance that noise alone scores this high at one 5 s look (−log10), calibrated on 14 days of noise (1-15 March 2014) or, for 3275 and 3315, 8 March records outside 00:15-02:00; band energy 8-40 Hz in 30 s; impact-pattern correlation from the whitened waveform; combined = Fisher sum of five scores. Calibration: USGS M≥3.5 earthquake arrival windows and faulty records removed, tail fix of 6 Nov (empirical ranks up to the noise maximum). Window p: chance that noise alone gives a maximum this high over the looks in the window. Each score uses 30 s windows and a ±20 s lag, so one short sound gives a flat-topped step about 70 s wide. Green line: raw 8-60 Hz power in 1 s steps (dB over the record median). Orange ticks: events noted earlier at this site (label only). Top strip: expected arrival for each nearest arc latitude (0.25° bins).
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